LINJI WANG

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SUMMARY

Ph.D. student in Computer Science (AI and Robotics) focused on curriculum learning and reinforcement learning for robot navigation and locomotion; ongoing work on humanoid robots.

First author of two IROS 2025 papers (GACL, with Peter Stone; Reward Training Wheels); co-author of a third IROS 2025 paper and an IEEE RA-L 2025 article. Team won 1st place in the simulation phase of the 2025 BARN Challenge.

Engineering experience at AWS building production-grade statistical testing and monitoring systems that cut regression analysis time from 8 hours to 15 minutes.

EDUCATION

George Mason University — Fairfax, VA
PhD in Computer Science, Specialization in AI and Robotics — *Sep 2023 – Present*

- Research: Curriculum Learning and Reinforcement Learning for Robotics, advised by Dr. Xuesu Xiao (RobotiXX Lab)

Carnegie Mellon University — Pittsburgh, PA
MSc in Mechanical Engineering, GPA: 3.94/4.0 — *Sep 2021 – May 2023*

University of Cincinnati — Cincinnati, OH
BS in Mechanical Engineering — *Sep 2016 – May 2021*

RESEARCH EXPERIENCE

RobotiXX Lab, George Mason University — Fairfax, VA
Graduate Research Assistant, Curriculum Learning for Robotics — *Aug 2023 – Present*

- Developed GACL (first author, IROS 2025, with Peter Stone), a grounded adaptive curriculum framework: +6.8% success on wheeled navigation in constrained environments, +6.1% on quadruped locomotion in confined 3D spaces vs. state of the art.
- Designed Reward Training Wheels (first author, IROS 2025), teacher-adapted auxiliary rewards: +122.62% off-road mobility, 3× faster training, and 5/5 vs. 2/5 success in physical off-road robot trials vs. expert-designed rewards.
- Contributed to Decremental Dynamics Planning (IROS 2025): navigation system won 1st place in the simulation phase of the 2025 BARN Challenge.
- Trained massively parallel RL policies (PPO, SAC) in IsaacGym across wheeled, quadruped, and off-road platforms; extending curriculum learning to humanoid robots.

Computational Engineering and Robotics Lab, CMU — Pittsburgh, PA
Research Assistant: 3D AR Scene Inpainting via Deep Learning — *Jan 2022 – May 2023*

- Developed an end-to-end deep learning pipeline for 3D AR scene inpainting, achieving 92% accuracy in scene completion.
- Fine-tuned a GAN model for image inpainting, improving texture realism by 35% over baseline methods.

INDUSTRY EXPERIENCE

Amazon Web Services (AWS) — Bellevue, WA
Software Development Engineer Intern - RDS Proxy Team — *May 2025 – Aug 2025*

- Built a Streamlit performance-analysis suite (8 interactive views) cutting regression analysis time from 8 hours to 15 minutes.
- Developed a statistical regression testing framework (Welch's t-test, power analysis, Bonferroni correction) for high-confidence detection.
- Implemented adaptive sampling with Thompson Sampling and Bayesian optimization, improving test reliability from 47% to 90% and reducing false positives.

PUBLICATIONS

Wang, L., Xu, Z., Stone, P., Xiao, X. "GACL: Grounded Adaptive Curriculum Learning" — IROS 2025
Wang, L., Xu, T., Lu, Y., Xiao, X. "Reward Training Wheels: Adaptive Auxiliary Rewards" — IROS 2025
Lu, Y., Xu, T., Wang, L., Hawes, N., Xiao, X. "Decremental Dynamics Planning" — IROS 2025
Zhao, C., Li, Y., Jian, Y., Xu, J., Wang, L., et al. "II-NVM: Normal Vector-Assisted Mapping" — IEEE RA-L 2025

AWARDS

1st Place, Simulation Phase — 2025 BARN Challenge (Benchmark Autonomous Robot Navigation) — ICRA 2025

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Programming — George Mason University — Fall 2023
Teaching Assistant, Artificial Intelligence and Machine Learning — Carnegie Mellon University — Fall 2022
Teaching Assistant, System Dynamics, Fluid Dynamics & Engineering Models — University of Cincinnati — Fall 202

TECHNICAL SKILLS

Programming:	Python, C++, CUDA, Bash
Robot Learning:	PyTorch, PPO/SAC, Curriculum Learning, Reward Design, Sim-to-Real
Robotics & Simulation:	IsaacGym, MuJoCo, ROS; wheeled UGV, quadruped, off-road platforms
Engineering & Cloud:	AWS, Docker, Git/CI-CD, Statistical Analysis